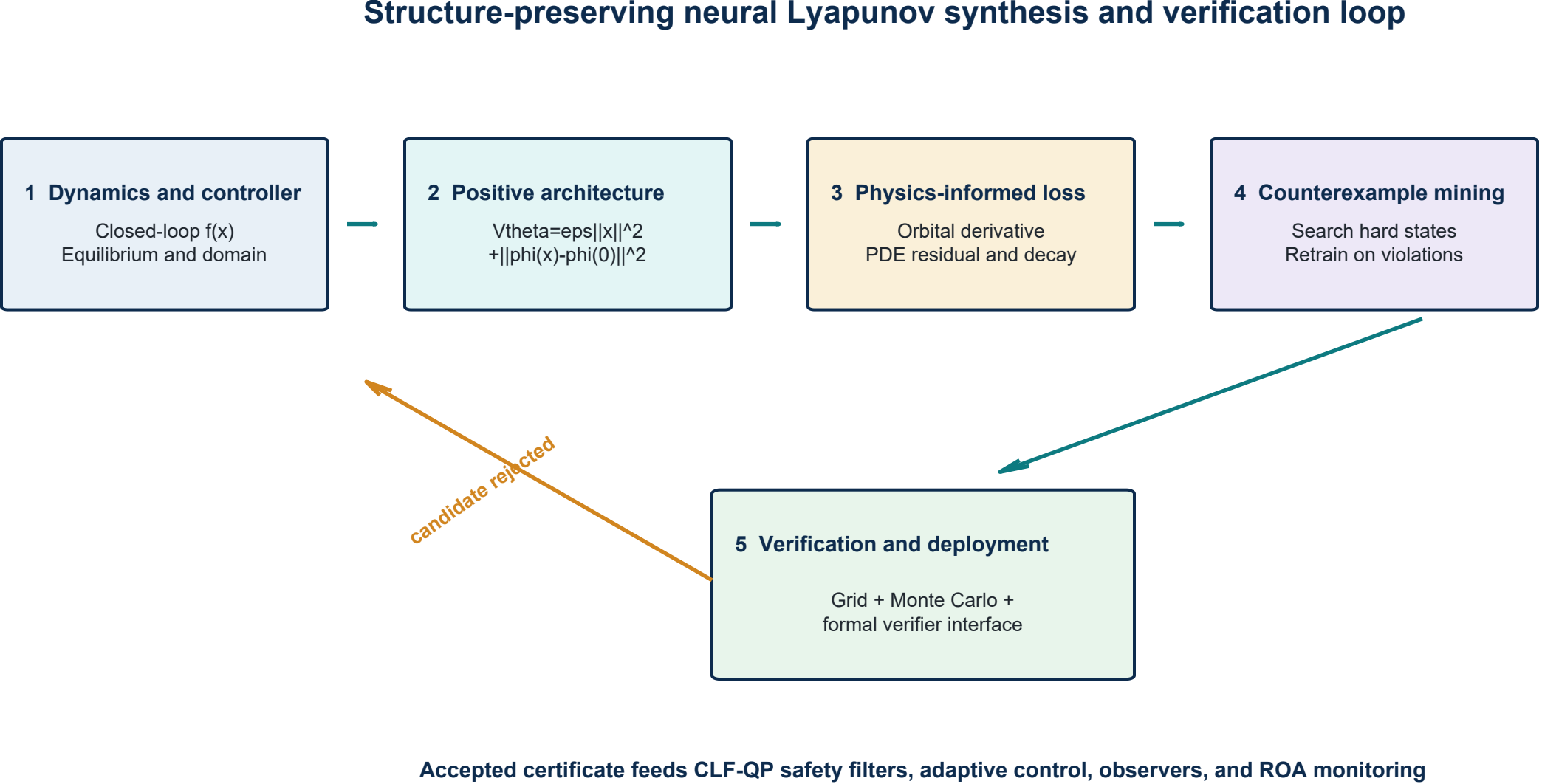
Structure-preserving neural Lyapunov synthesis and verification loop
1 Dynamics and controller
Closed-loop f(x)
Equilibrium and domain
2 Positive architecture
Vtheta=eps||x||^2
+||phi(x)-phi(0)||^2
3 Physics-informed loss
Orbital derivative
PDE residual and decay
4 Counterexample mining
Search hard states
Retrain on violations
candidate rejected
5 Verification and deployment
Grid + Monte Carlo +
formal verifier interface
Accepted certificate feeds CLF-QP safety filters, adaptive control, observers, and ROA monitoring